ARBITRARY-DEPTH EXPONENTIAL HOMFLY-PT FIBERS OVER QUIVER POINT-COUNTING POLYNOMIALS OF PLABIC LINKS AND KNOTS

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with the Galashin–Lam problem on whether the recursive $\mathbb{F}_q$ quiver point-counting polynomial of a simple plabic graph determines the full two-variable HOMFLY-PT polynomial of its associated plabic link, we construct explicit infinite families of connected, simple, non-reduced plabic realizations of trees and corona-lifted trees of bounded degree. For every integer jet depth $r \geq 1$ and word length $m \geq 1$, we obtain 2^m connected simple plabic realizations on $N = 2n(r)m + 8$ vertices ($n(r) \leq 37r^2$) whose associated links are single-component knots sharing the same normalized quiver point-counting polynomial, the same Alexander specialization, the same first r top-a HOMFLY-PT layers, and all a-derivatives of order less than r at $a = 1$, while producing 2^m pairwise distinct two-variable HOMFLY-PT polynomials ($\log_2 |\text{fiber}| \geq (N - 8)/(74r^2)$).

1. Main Mathematical Results

Theorem 1.1 (Arbitrary-depth exponential HOMFLY-PT fibers for plabic knots)**.** *For every pair of integers $r, m \geq 1$, there exists an explicit family of 2^m connected trees T^+_ϵ ($\epsilon \in \{0,1\}^m$) of maximum degree at most six on $N = 2n(r)m + 8$ vertices ($n(r) \leq 37r^2$) whose polygon-cell plabic disk realizations are connected, simple, non-reduced, and yield single-component knots such that:*

(1) *All 2^m members have identical normalized quiver point-counting polynomial $R(q)$, identical Alexander specialization, and identical HOMFLY-PT derivatives of all orders $< r$ at $a = 1$;*
(2) *Their r-th derivatives at $a = 1$, and hence their full two-variable HOMFLY-PT polynomials $P(a, z)$, are pairwise distinct, giving $\log_2 |\text{fiber}| \geq (N - 8)/(74r^2)$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001675 in the domain of algebraic combinatorics, cluster algebras, and knot polynomials adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.
(c) *Resolved scope:* Explicit connected simple non-reduced plabic link and knot families of arbitrary jet depth r >= 1 where a single quiver point-counting and Alexander fiber contains 2^m distinct two-variable HOMFLY-PT polynomials.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Reduced simple plabic graphs and mutation-equivalence invariance of full two-variable HOMFLY-PT polynomials

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References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 20001675 (algebraic combinatorics, cluster algebras, and knot polynomials), archived in the accompanying repository package `results/20001675/`.